



Department of Cyber Security
Air University Islamabad
Mid-term Examination: Spring 2025

Student ID: 242292

Subject: Digital Logic Design
Course Code: EE-123
Class: BSCYSev
Semester: Spring 2025
Section(s): A,B

Total Marks: 100
Date: 15 April, 2025
Time: 11:30 to 1:30
Max. Time Allowed: 2 Hrs

FM(s) Name: Dr. Muhammad Awais Shahid,
Engr. Sharoze Sohail

Special Instructions:

- Paper contains 04 questions spanned over 02 pages. Make sure that you have got the complete question paper.
- Vague or badly motivated answers will be considered **wrong**.
- Lead Pencils can **NOT** be used for solving the exam. **Calculators are not Allowed.**
- In case of any ambiguity, please make use of your Assumption "Explicitly"
- It is mandatory to attach question paper with the answer book.

Question # 1:

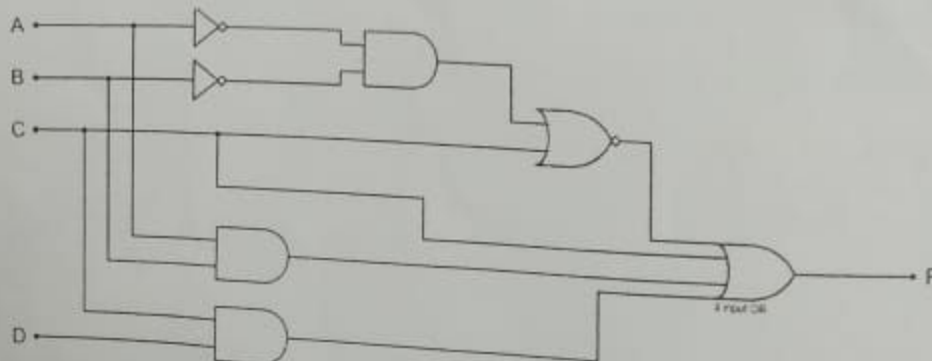
CLO1/PLO1(30)

- (a) Perform the subtraction on the given unsigned binary numbers using the 2's complement of the subtrahend. $110000 - 10101$
- (b) Explain the process of BCD addition by performing the addition of 548 and 679. Convert the answer back to decimal.
- (c) Explain Gray Coding and write down the gray code table for three inputs.

Question # 2:

CLO2/PLO2(30)

- (a) Interpret the Boolean Expression F for the logic diagram given in figure and convert the obtained function into simplified form of 3 literals.



- (b) Illustrate and express $F = xy + yz'$ the given standard form in Sum of Minterms form using algebraic manipulation. Also write the expression in maxterm designation.

(c) Show and simplify the Boolean function: $F = AB + (AC)' + AB'C(AB + C)$.

Question # 3:

CLO2/PLO2(40)

(a) Show $F(A, B, C, D) = \Sigma(1, 3, 8, 10, 15)$ with don't care conditions $d(A, B, C, D) = \Sigma(0, 2, 9)$.

i. Using K-Map, find the simplified Sum of Products form of F.

ii. Using K-Map, find the simplified Product of Sums form of F.

iii. Which of POS and SOP is more simplified? Draw the simplified logic diagram for F.

(b) Explain the given function and implement it with all-NOR circuit: $F = wx' + y'z' + w'yz'$

(c) Show and prove $x \oplus 1 = x'$ and $x \oplus 0 = x$.

(d) Demonstrate and simplify the following Boolean functions by first finding the essential prime implicants:

$$F(A, B, C, D) = \Sigma(1, 3, 4, 5, 10, 11, 12, 13, 14, 15)$$

Best Of Luck ☺